HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.

TOÁN RỜI RẠC

ĐẠI HỌC
BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
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DISCRETE MATHEMATIC

Minimum Spanning Tree

ONE LOVE. ONE FUTURE.

Part 1: Combinatorial theory

Part 2: Graph theory

Content: part 2

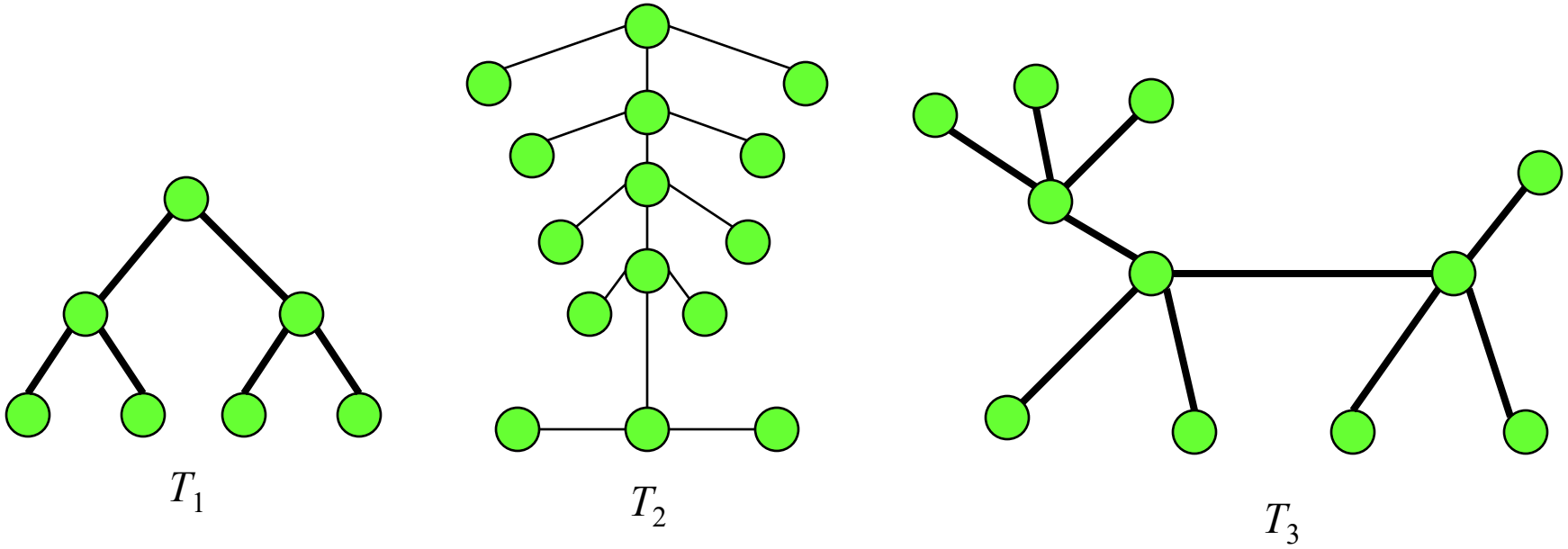
- Chapter 1. Definitions
- Chapter 2. Graph representation
- Chapter 3. Graph traversal
- **Chapter 4. Tree and Minimum Spanning Tree Problem**
- Chapter 5. Shortest path problem
- Chapter 6. Max-Flow

Chapter 4. Tree and Minimum Spanning Tree problem

- Definitions
- Spanning Tree
- Minimum Spanning Tree Problem

Definitions

- Tree is an undirected, connected graph, and contains no cycle
- Forest is an undirected graph containing no cycle
- Each connected component of a forest is a tree



Forest *F* has 3 trees T_1, T_2, T_3

Definitions

Theorem. Given an undirected graph $G = (V,E)$, the following conditions are equivalent:

(1) G is a connected graph with no cycles. (Thus G is a tree by the above definition).

(2) For every two vertices $u, v \in V$, there exists exactly one simple path from u to v.

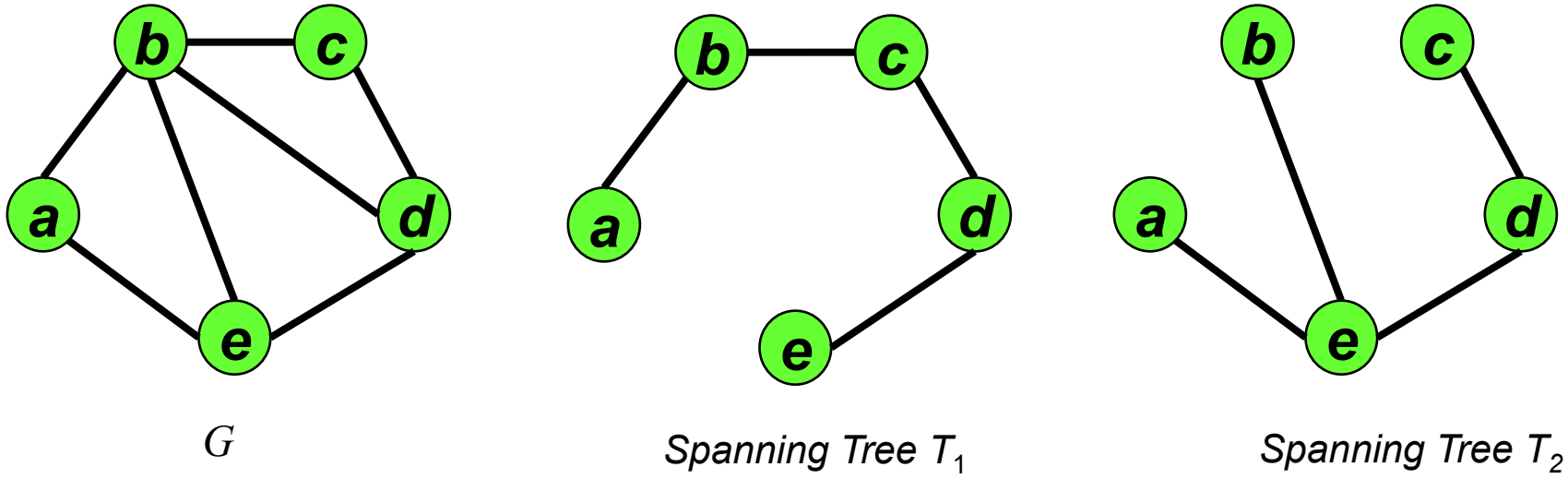
(3) G is connected, and removing any edge from G disconnects it (each edge of G is a bridge).

(4) G has no cycles, and adding any edge to G gives rise to a cycle. (Thus G is a maximal acyclic graph).

(5) G is connected and $|E| = |V| - 1$.

Spanning Trees

- Definition. *Let G=(V,E) is an undirected connected graph. A Tree T=(V,F) having F⊆ E called a spanning tree of G.*

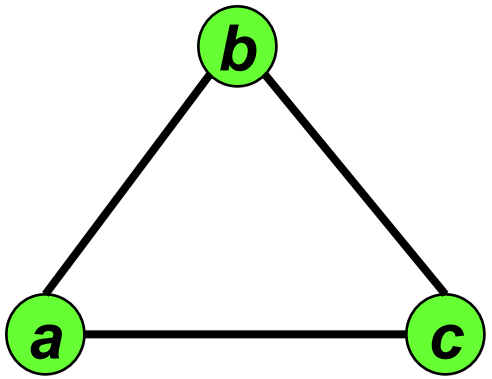


G *Spanning Tree T_1* *Spanning Tree T_2*

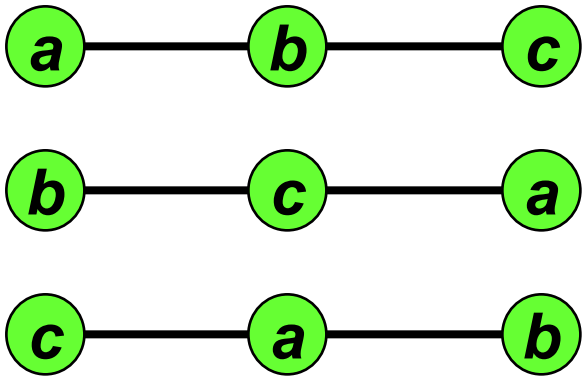
Graph G and 2 spanning trees T_1 and T_2

Spanning trees

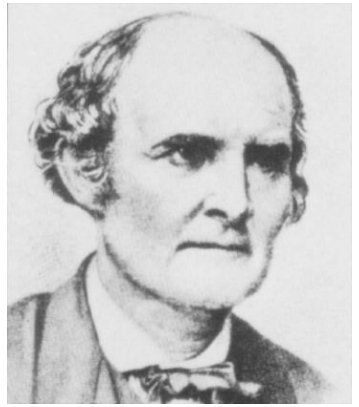
- **Theorem 2 (Cayley).** *Number of spanning trees of* K_n *is* n^{n-2}



K_3



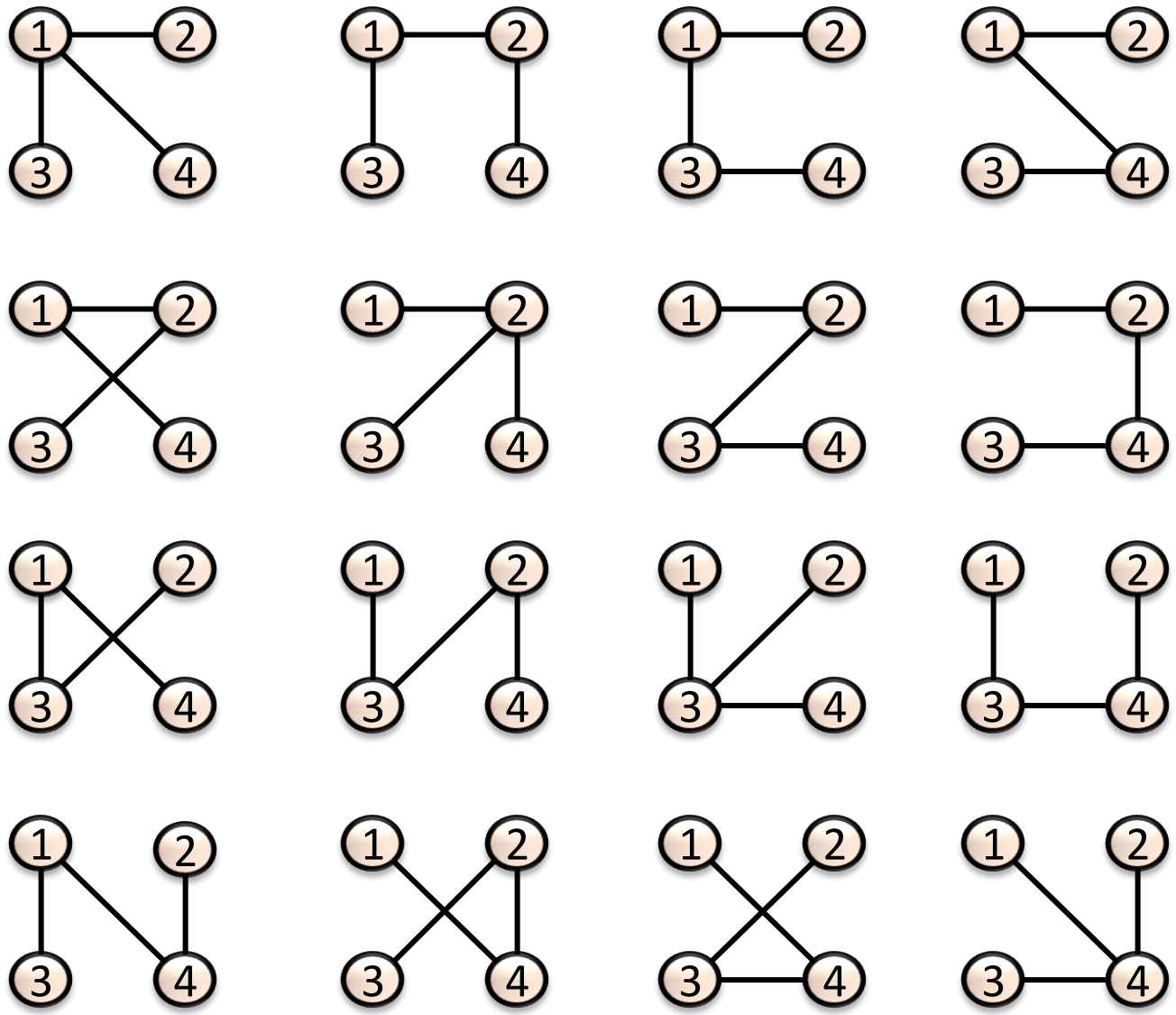
3 spanning trees of K_3



Arthur Cayley
(1821 – 1895)

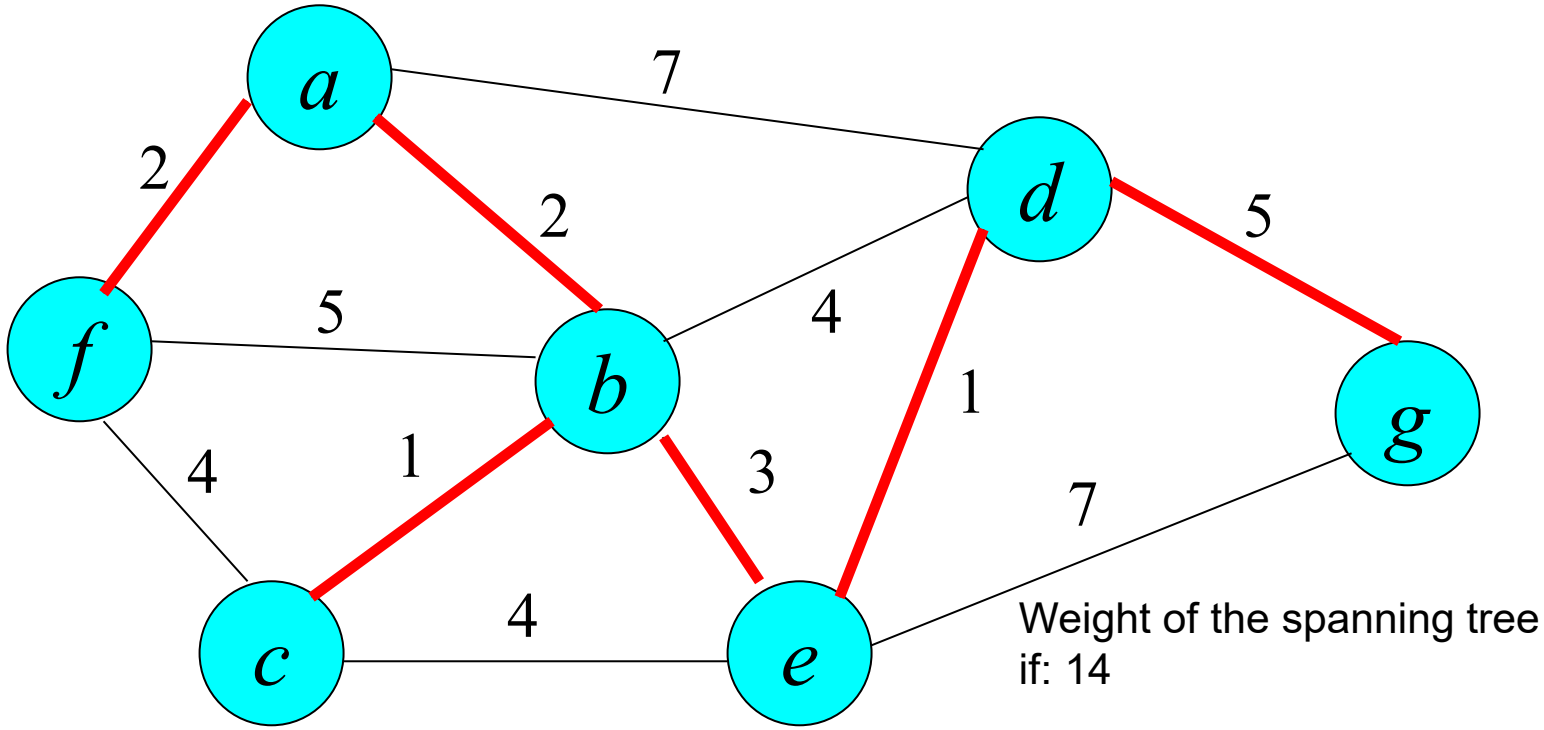
Spanning trees

- **Theorem 2 (Cayley).** *Number of spanning trees of* K_n *is* n^{n-2}



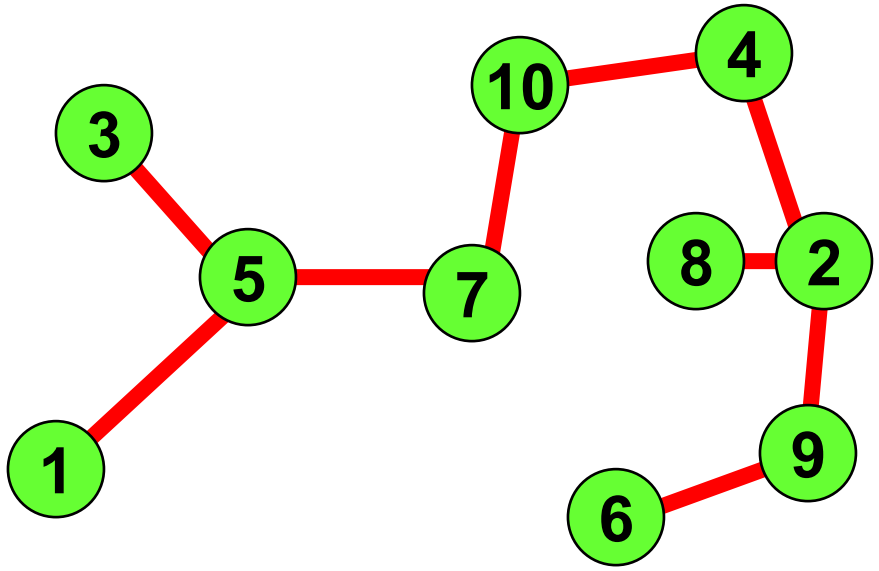
Minimum Spanning Tree problem

- **Problem:** Given an undirected connected graph $G=(V,E)$ in which $c(e)$ is the weight of edge e ($e \in E$). The weight of a tree is the sum of weights of its edges. Find the spanning tree of G having the smallest weight



Minimum Spanning Tree problem (applications)

- Company AT&T want to setup a network connecting *n* customers. Connection cost between *i* and *j* is c_{ij}. Find the network having the smallest weight?



Minimum Spanning Tree problem (applications)

Railway construction

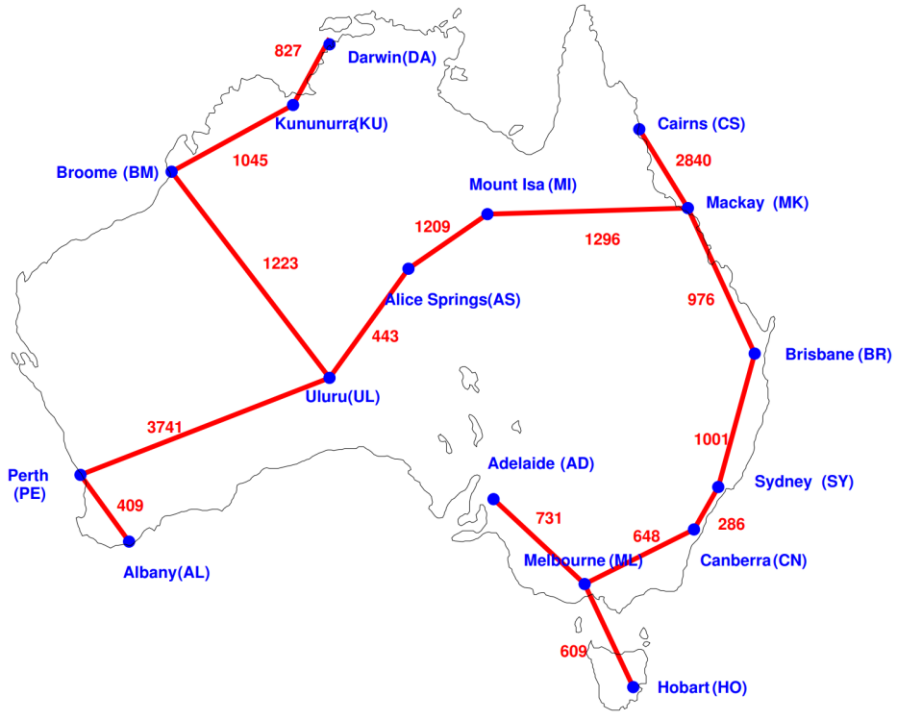
- Suppose we want to build a railway system that connects *n* cities so that passengers can travel between any two cities and the total cost of construction must be minimal.

- It is clear that it corresponds to a graph where vertices are the cities and the edges are the railroads connecting the cities, the length on each edge is the cost of building a railway connecting the two cities.

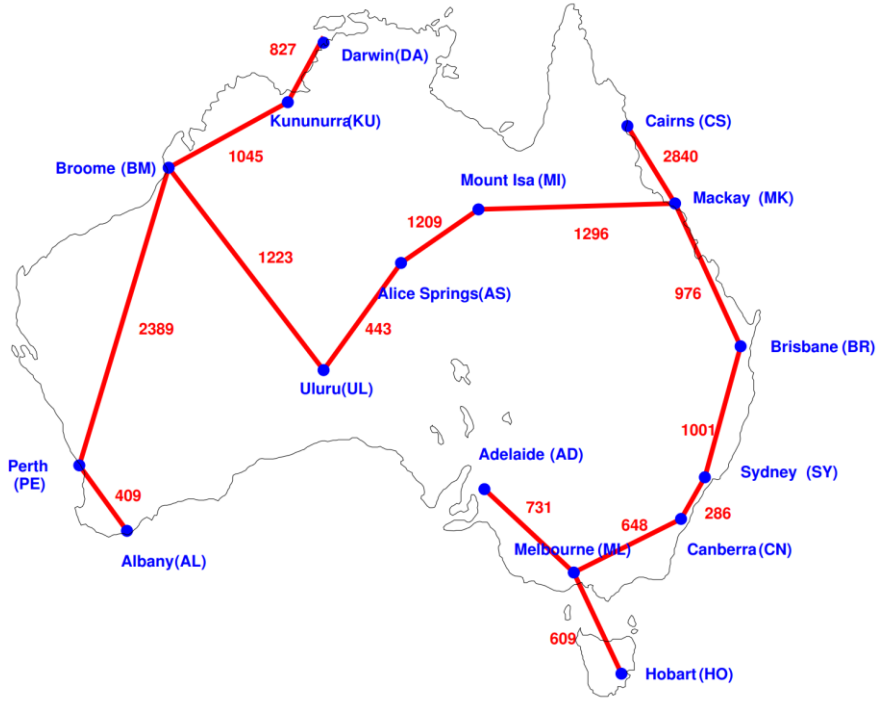
- Therefore, it leads to the problem of finding the smallest spanning tree on a complete graph K_n

Minimum Spanning Tree problem (applications)

Railway construction



Length: 17284



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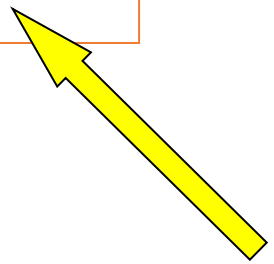
Generic Algorithm Schema for Minimum Spanning Tree problem

```
Generic-MST(G, c)
    A = ∅
    //Invariant: A subset of edges of some MST
    while A is not a spanning tree do
            select a safe edge (u, v) of A
            A = A ∪{(u, v)}
            // A is still a subset of edges of some MST
    return A
```



Find the smallest weight edge for inserting into A, that does not create a cycle

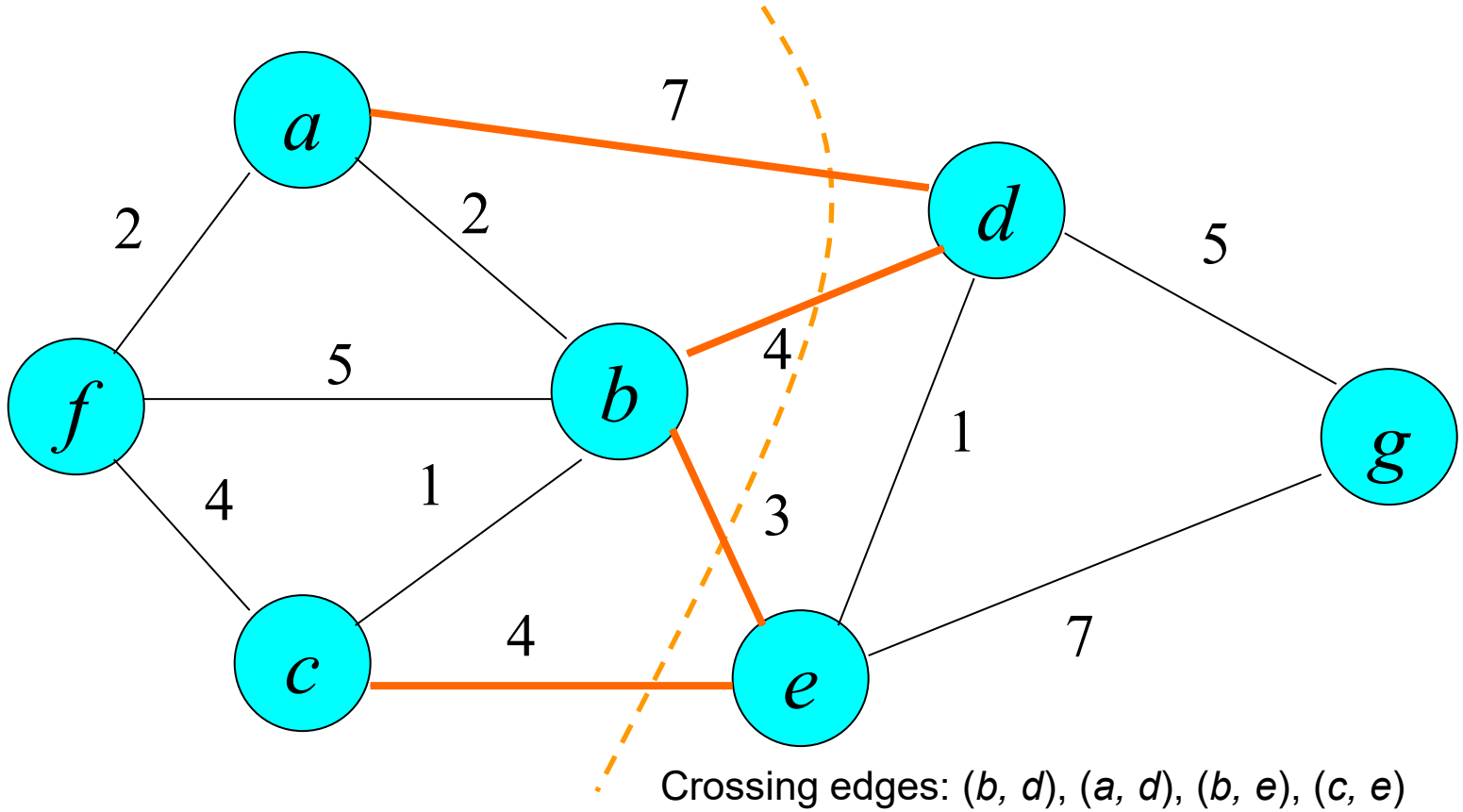
How to find a safe edge?

Cuts

- A cut $(S, V - S)$ is a partition of V into 2 subset S and $V - S$. An edge e is called a crossing edge $(S, V - S)$ if one endpoint belongs to S and the other belongs to $V - S$.

Cuts

- **A cut** of $G = (V, E)$ is a partition of V into $(S, V - S)$.
- **Example** A cut $(S, V\text{-}S)$: $S = \{a, b, c, f\}$, $V - S = \{e, d, g\}$



Cuts

- A cut $(S, V-S)$ is a partition of V into 2 subset S and $V-S$. An edge e is called a crossing edge $(S, V-S)$ if one endpoint belongs to S and the other belongs to $V-S$.
- Suppose A is a subset of edges of G. A cut $(S, V-S)$ is called *compatible* with A if no edge of A is a crossing edge of the cut

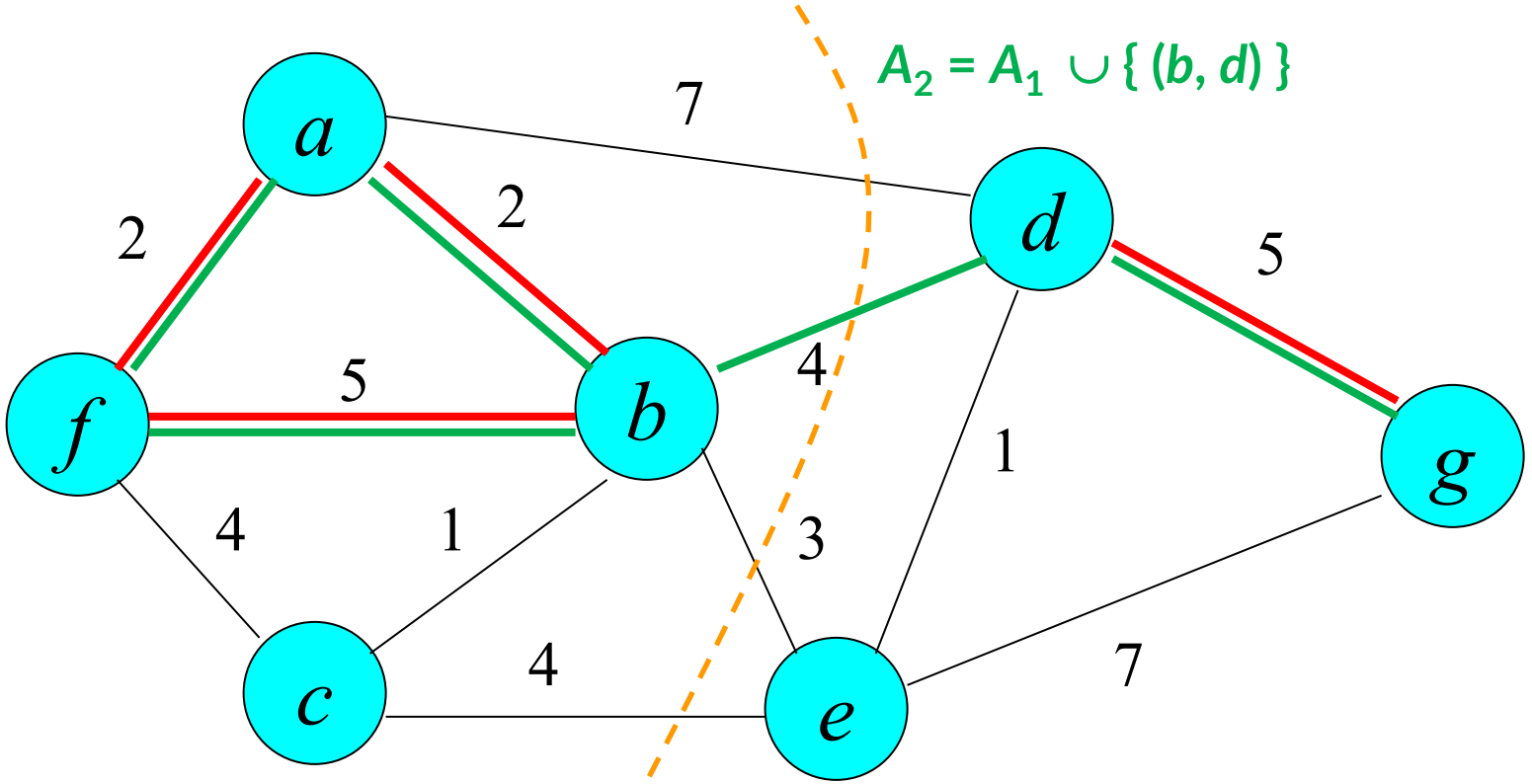
Cuts

Example. Cut $(S, V\text{-}S)$: $S = \{a, b, c, f\}$

Edges:

$A_1 = \{ (a, b), (d, g), (f, b), (a, f) \}$

$A_2 = A_1 \cup \{ (b, d) \}$

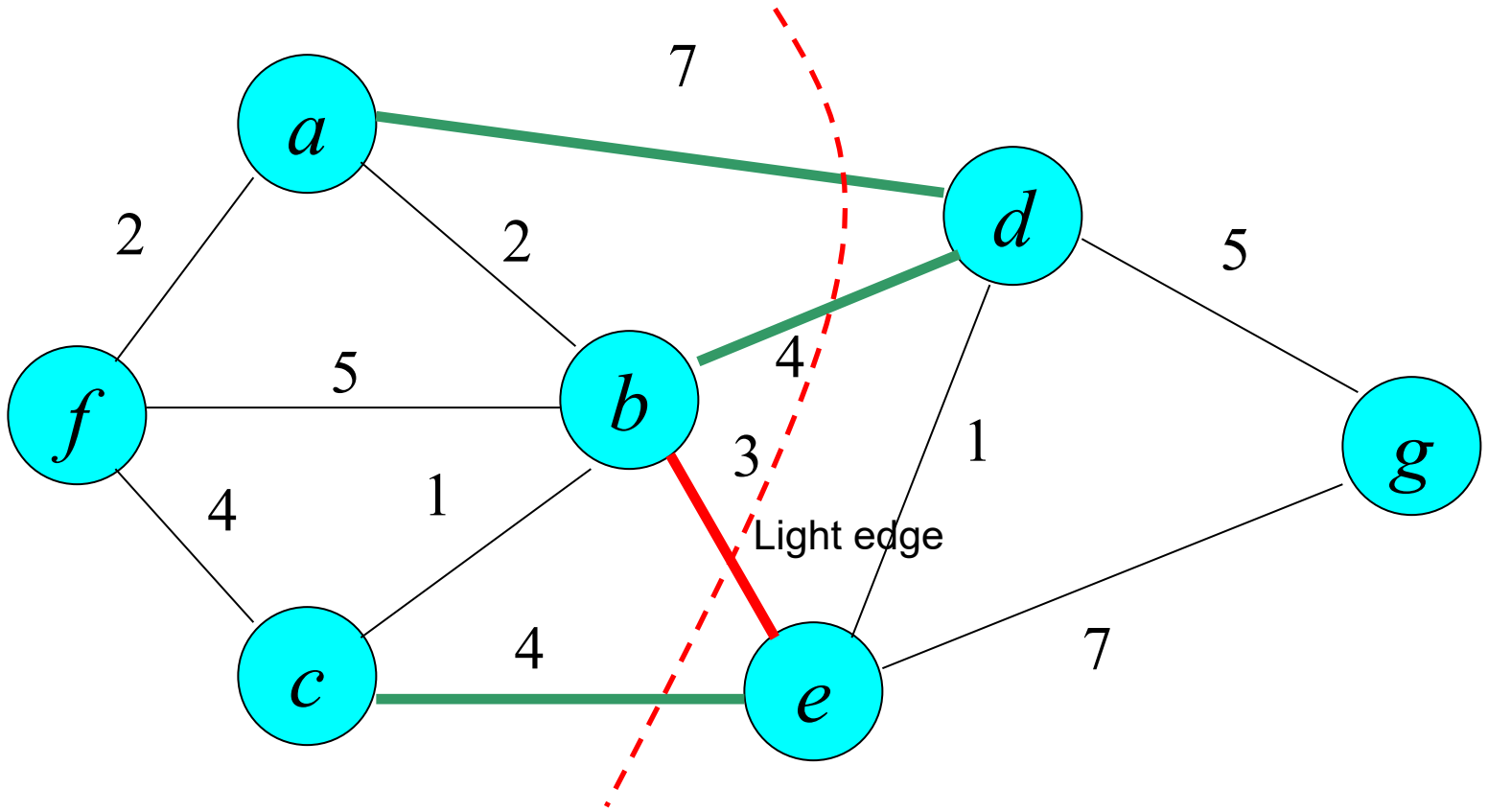


$(S, V - S)$ is compatible with A_1 and Not compatible with A_2

Light edge

- Light edge is an edge having the smallest weight among crossing edges of the cut

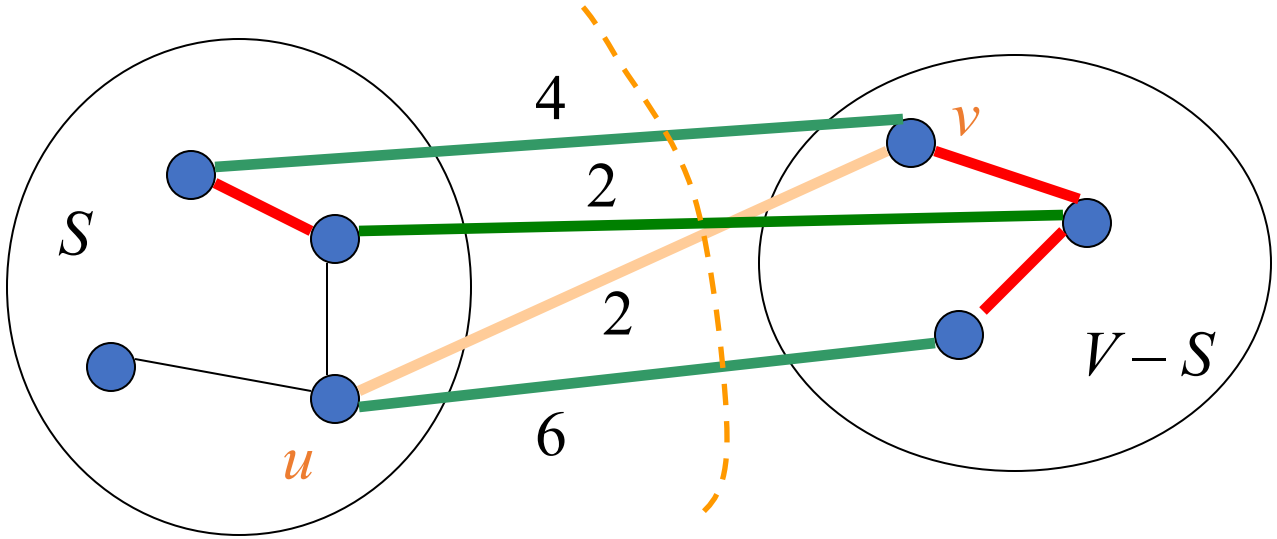
Example. Cut $(S, V\text{-}S)$: $S = \{a, b, c, f\}$



Edge (b, e) has weight 3, "smaller" than other crossing edges (a, d), (b, d), và (c, e).

Safe and light edge

- **Theorem.** Suppose $(S, V - S)$ is a cut of $G=(V, E)$ which is compatible with a subset A of E, and A is a subset of edges of a MST of G. Denote (u, v) a light and crossing edge of $(S, V - S)$. Hence, (u, v) is a safe edge of A;

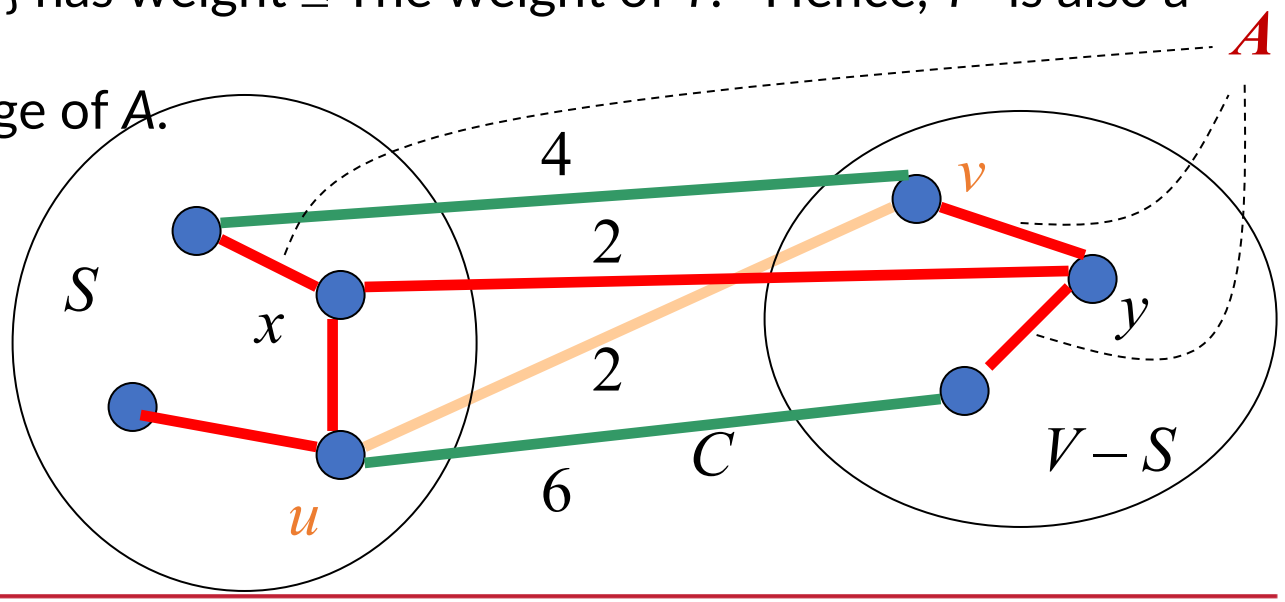


A consists of red edges.

Safe and light edge

- **We show.** (u, v) is a ***safe*** edge for A; it means: $A \cup \{(u, v)\}$ is also a subset of edges of some MST of G.
- Suppose T is a MST (red edges) containing A.

  Suppose the light edge $(u, v) \notin T$. We have
  - $T \cup \{ (u, v) \}$ contains a cycle
  - We can find an edge $(x, y) \in T$ which is a crossing edge of $(S, V - S)$.
  - The spanning tree $T' = T - \{ (x, y) \} \cup \{ (u, v) \}$ has weight $\leq$ The weight of T. Hence, T' is also a MST of G.
  - $A \cup \{ (u, v) \} \subseteq T'$, it means, (u, v) is a safe edge of A.



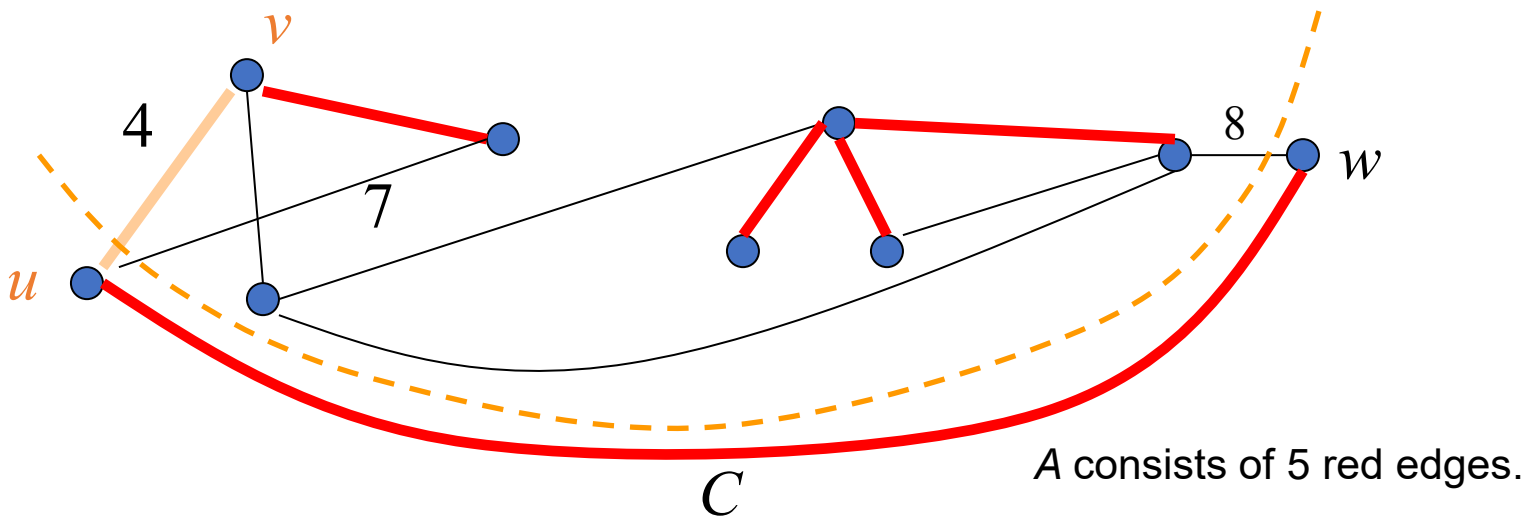
Safe and light edge

Consequence Let A be a subset of E and is a subset of the edges of some MST of $G=(V,E)$, and C is a connected component of the forest $F = (V, A)$. If (u, v) is a light edge connecting C with another connected component of F, then (u, v) is a safe edge for A.

Proof

Edge (u, v) is a light and crossing edge of the cut $(C, V - C)$ which is compatible with A.

With above theorem, edge (u, v) is safe for A.



Find safe edge

- Suppose *T* is a subset of the edges of some MST.

Kruskal Algorithm

- *T* is a forest.
- The selected safe edge for *T* is the edge having the smallest weight that connecting two connected components of *T*

Prim Algorithm

- *T* is a tree.
- The selected safe edge is the edge having the smallest weight and connecting a node of *T* and another node that is not in *T*.

Kruskal Algorithm

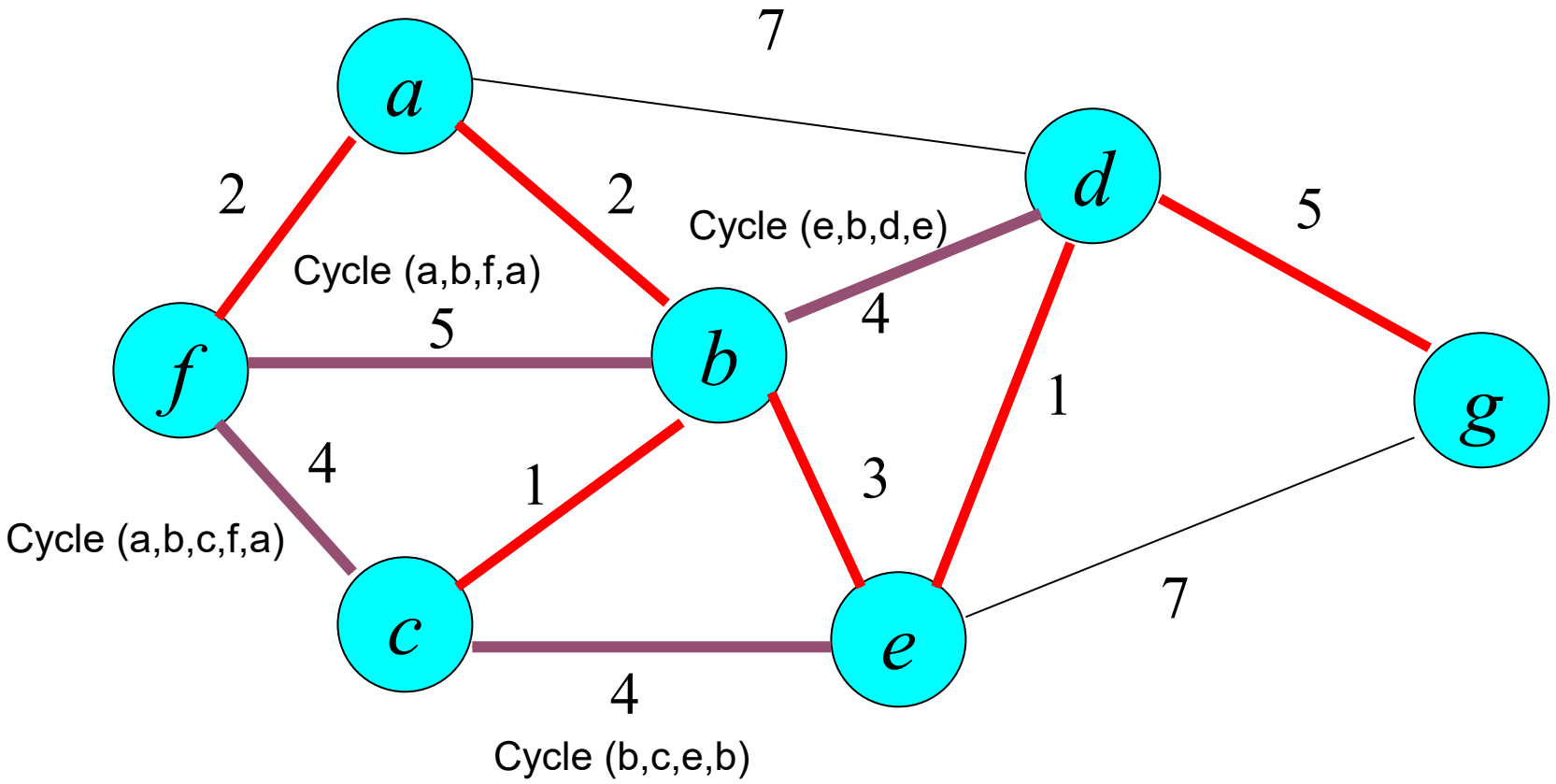
```
Generic-MST(G, c)
        T = ∅
       //Invariant: T is a subset of edges of some MST
        while T is not a spanning tree do {
            find a safe edge (u, v) for T
            T = T ∪{(u, v)}
            // T is still a subset of edges of some MST

        return T;
```

Kruskal Algorithm

- ★ *T* is a forest **(from an empty forest).**
- ★ The selected safe edge for *T* is the edge having the smallest weight that connecting two connected components of T

Kruskal Algorithm



Length of the spanning tree: 14

Kruskal Algorithm

```
Kruskal ( ){
    Sort m edges of G e_1, . . . , e_m in a non-decreasing order of weights;
    T = ∅;        // T: set of edges of the MST under construction
    for (i = 1; i <=m; i++)
         if (T ∪ {e_i} does not create any cycle) then
              T =  T ∪ {e_i};
}
```

Kruskal Algorithm

```
Kruskal ( ){
    Sort m edges of G e_1, . . . , e_m in a non-decreasing order of weights;
    T = ∅;        // T: set of edges of the MST under construction
    for (i = 1; i <=m; i++)
         if (T ∪ {e_i} does not create any cycle) then
              T =  T ∪ {e_i};
}
```

- Step 1. Sort edges.
  - Heap sort or merge sort take $O(m \log m)$
- Loop: Check if $T \cup \{ e_i \}$ contains a cycle?
  - Use DFS with time complexity of $O(m+n)$.
  - Time $O(m(m+n))$
- Total time complexity: $O(m \log m + m(m+n))$ in which n, m are number of nodes and edges of the given graph

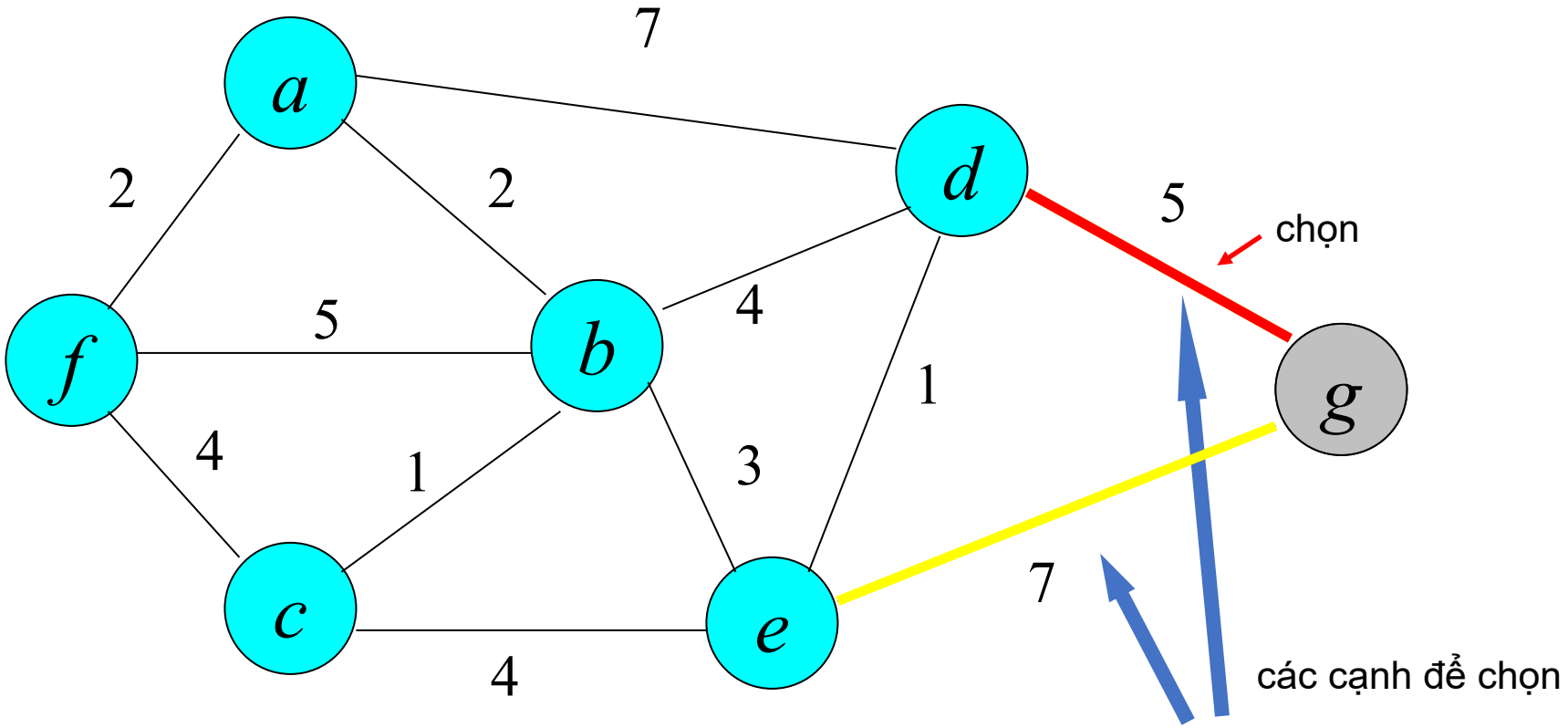
Prim Algorithm

```
Generic-MST(G, c)
       T = ∅
      //Invariant: T is a subset of edges of some MST
       while T is not a spanning tree do
           find a safe edge (u, v) for T
           T = T ∪{(u, v)}
           // T is still a subset of edges of some MST
       return T
```

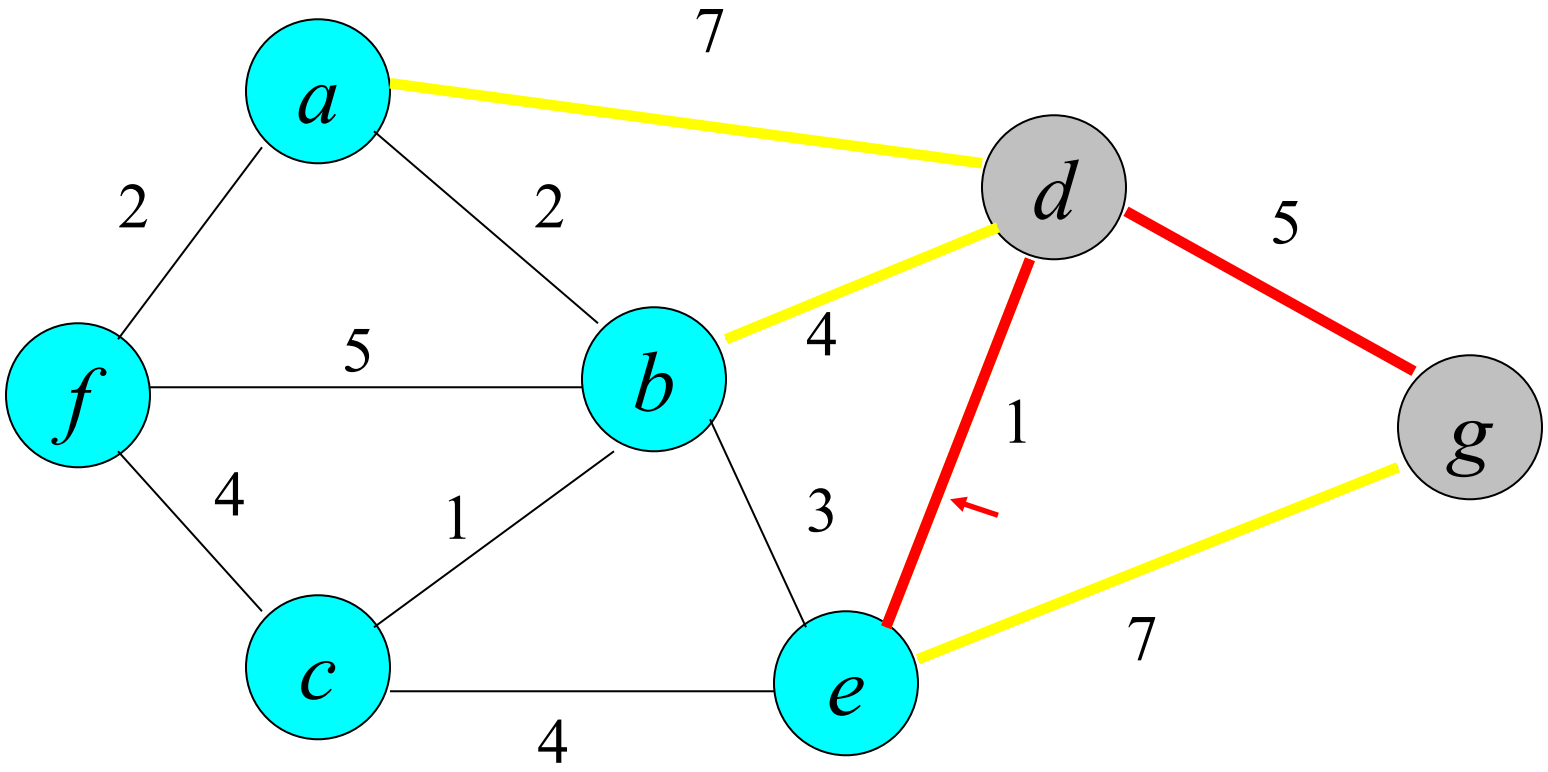
PRIM Algorithm

- ★ T is a tree (begin with a tree containing only one node)
- ★ The selected safe edge is the edge having the smallest weight and connecting a node of T and another node that is not in T.

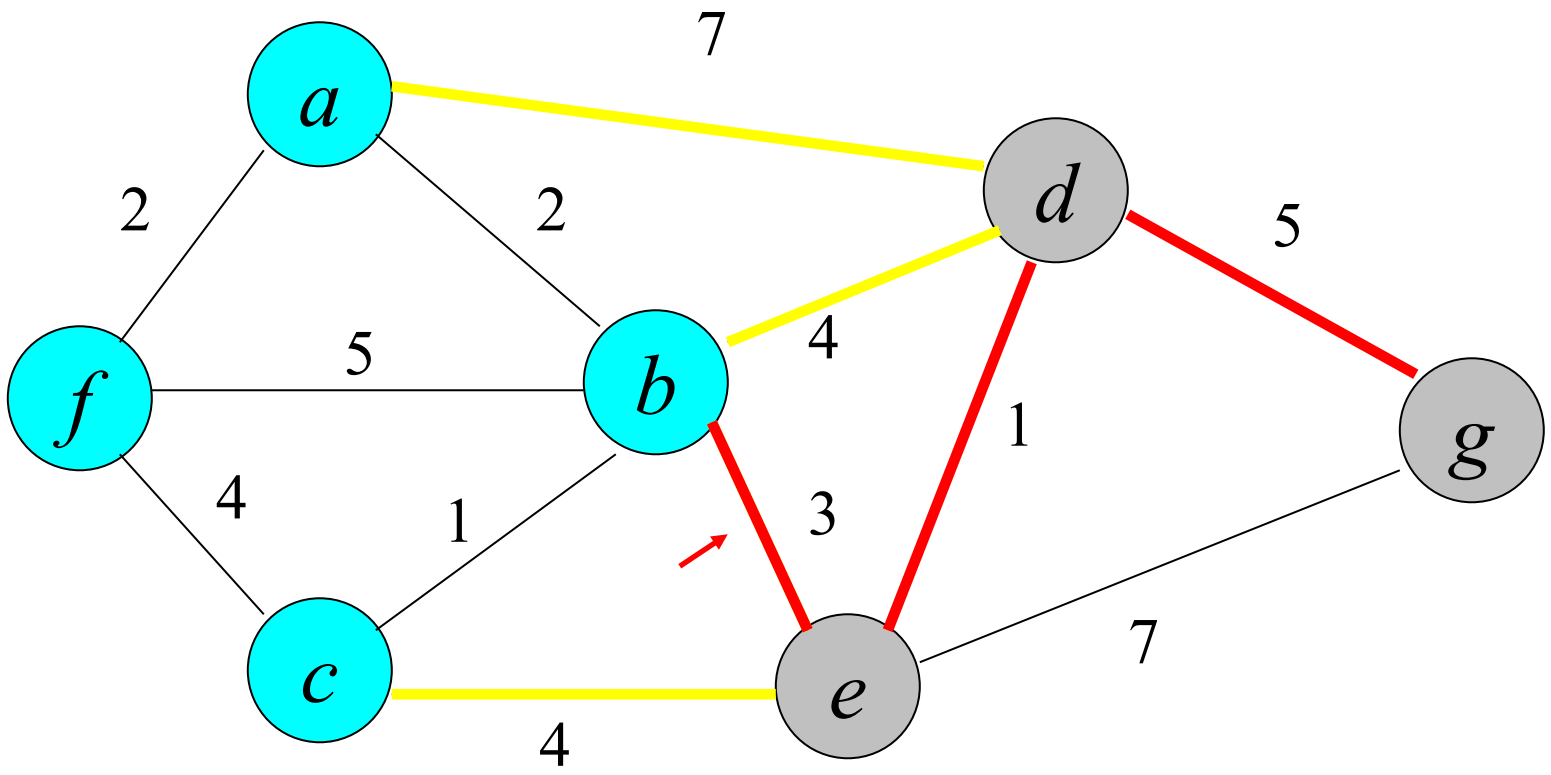
Prim Algorithm



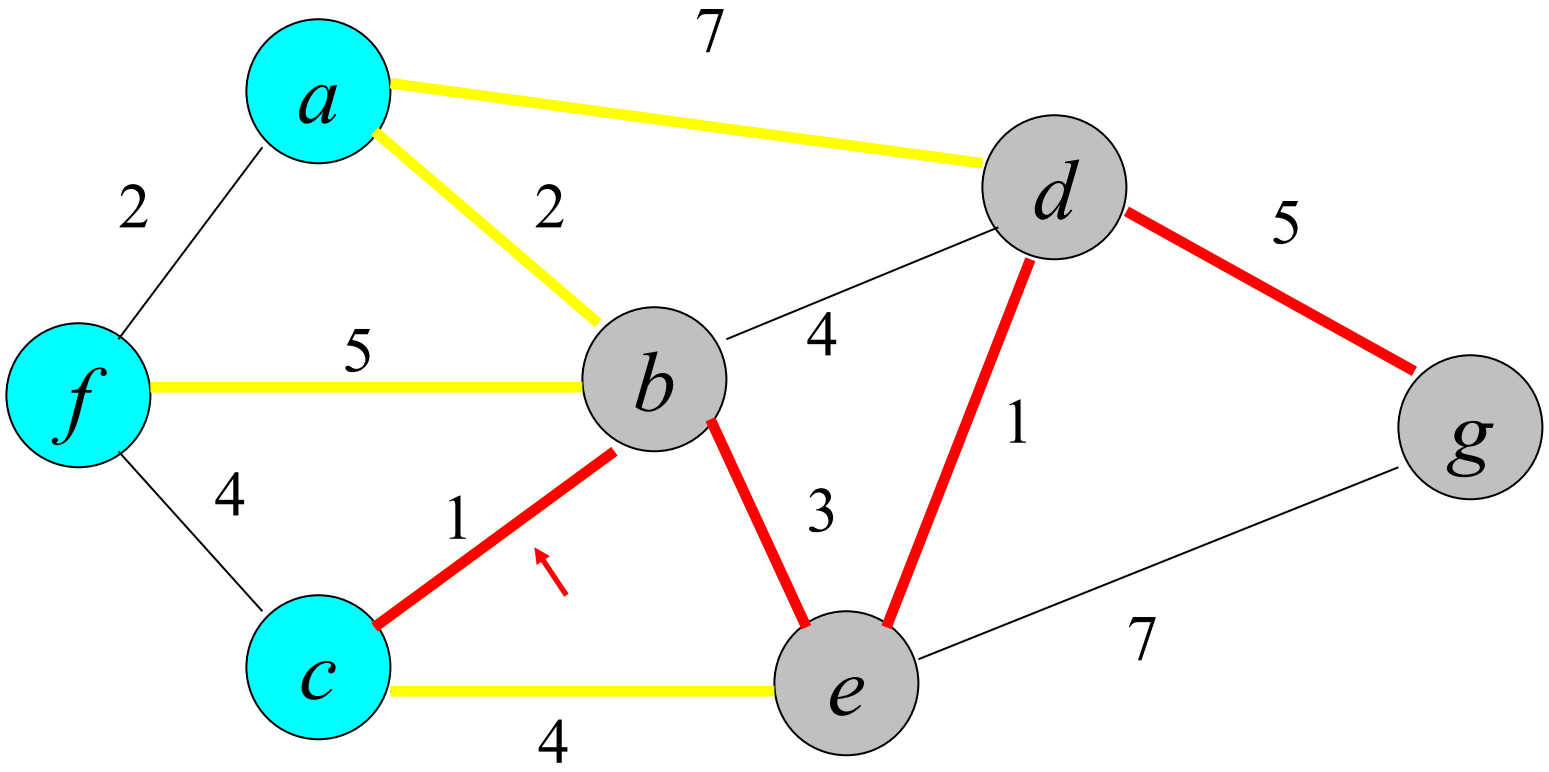
Prim Algorithm



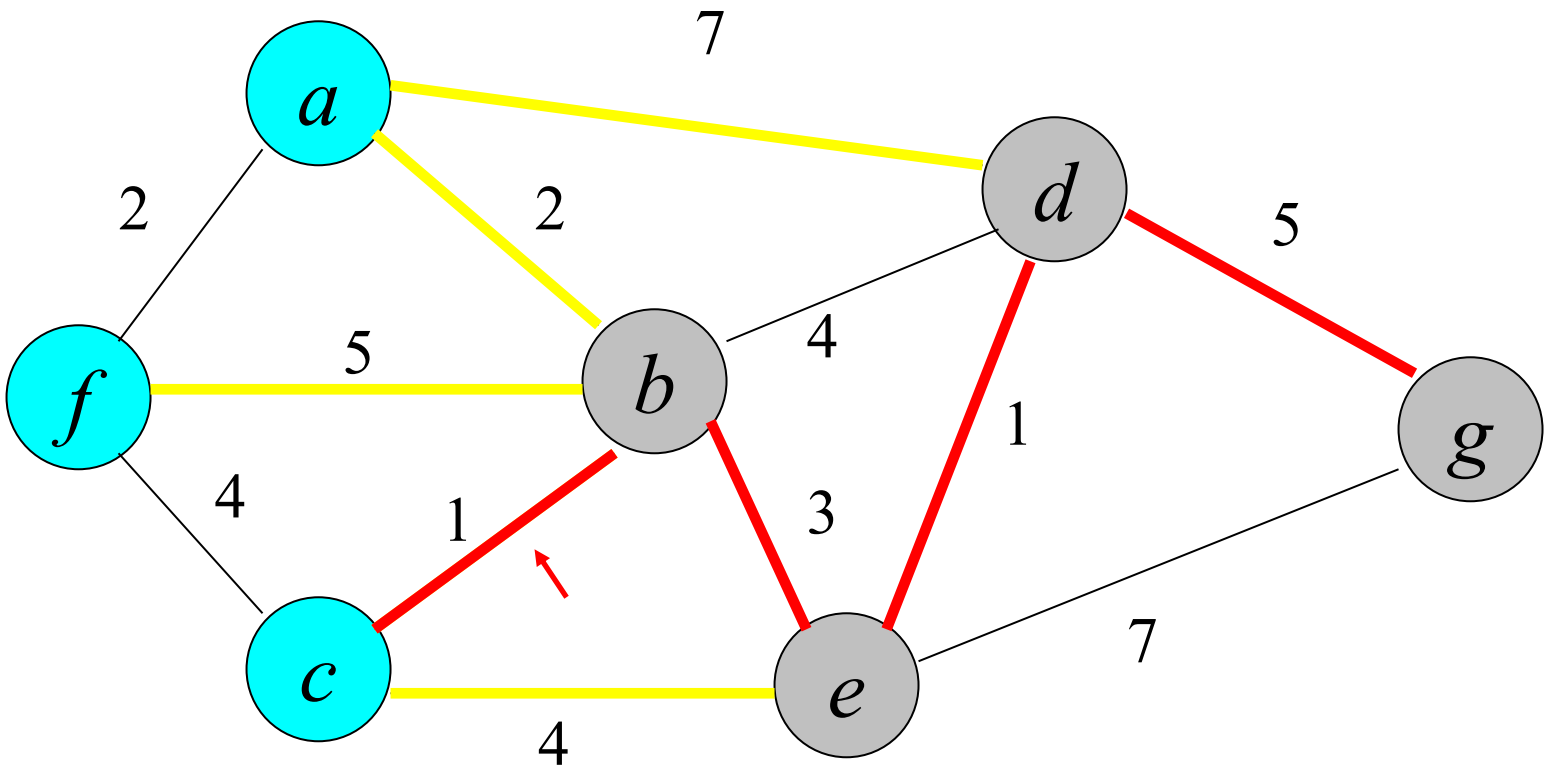
Prim Algorithm



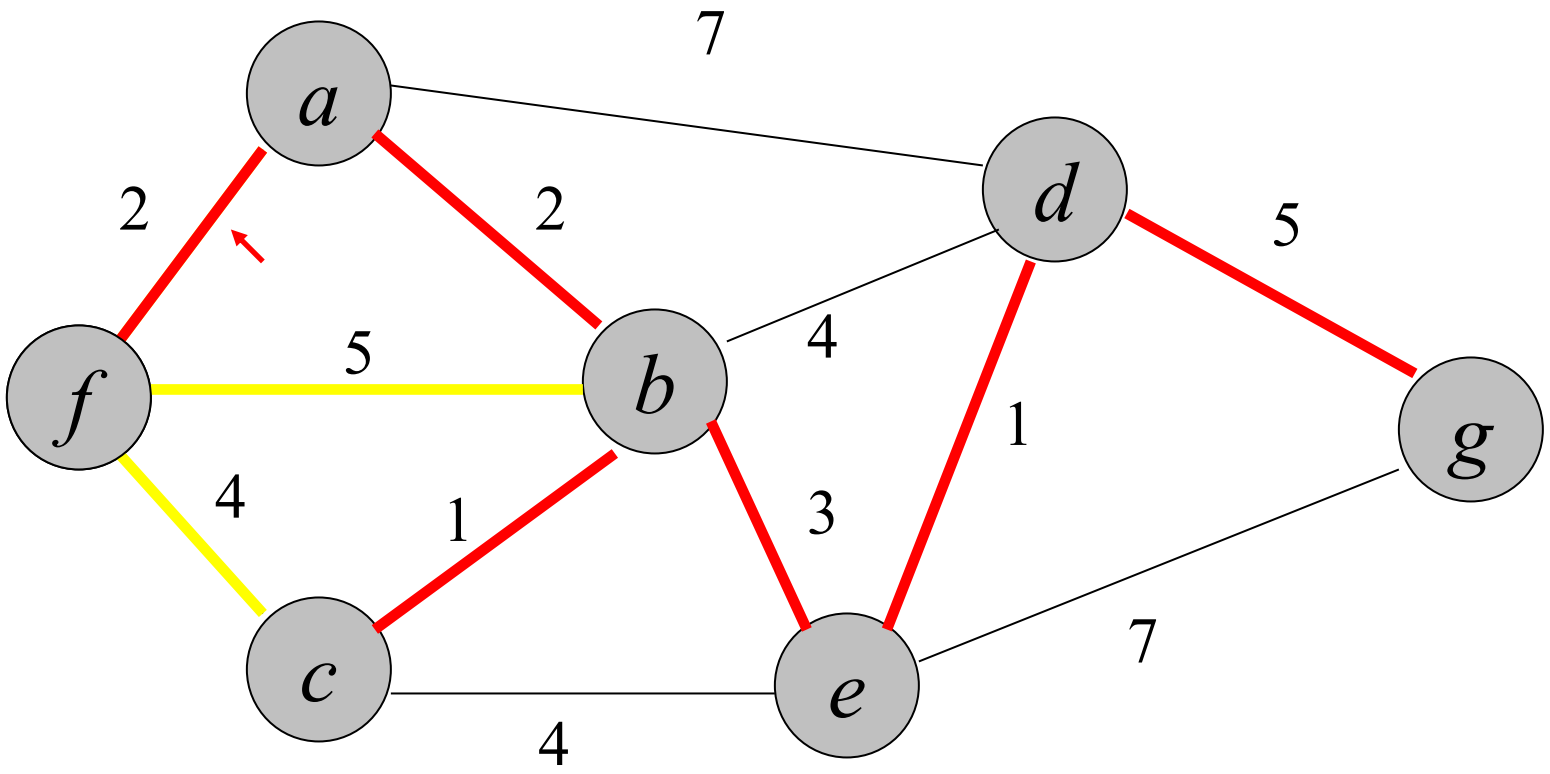
Prim Algorithm



Prim Algorithm



Prim Algorithm



CKNN gồm các cạnh: (*g*,*d*), (*d*,*e*), (*e*,*b*), (*b*,*c*), (*b*,*a*), (*a*,*f*)

Độ dài của CKNN: 14 5+1+3+1+2+2 = 14

Prim Algorithm

```
Prim(G, C){ //G: given graph; C: weight matrix
        select a node  r ∈V;
        Initialize T=(V(T), E(T)) with V(T)= {r} and E(T)=∅;
        while  V(T) < n  {
                (u, v) ← select  the smallest weight edge with u ∈ V(T) and  v∈V(G) – V(T)
                 E(T) ← E(T) ∪ { (u, v) };
                V(T) ← V(T) ∪ { v }
        }
}
```

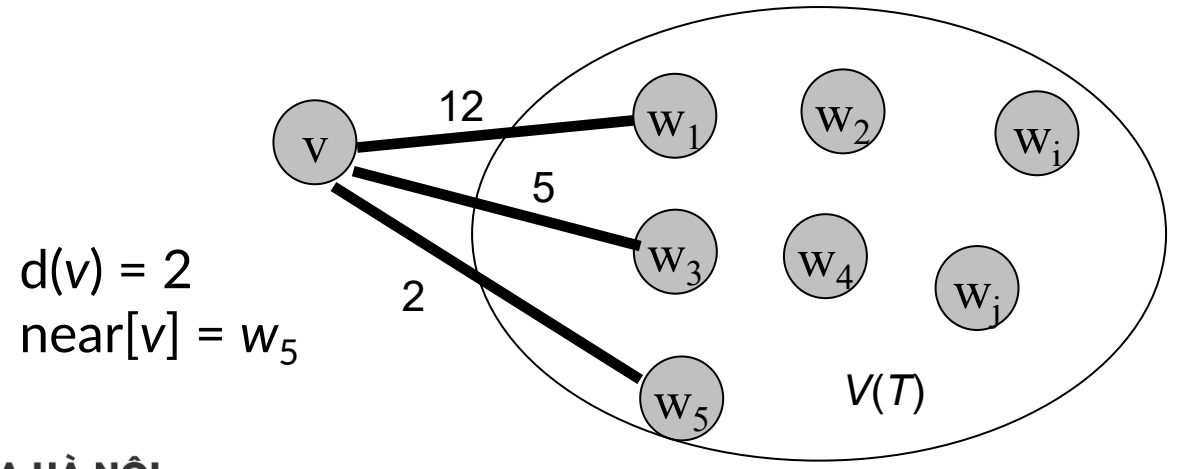
Tính đúng đắn suy từ hệ quả đã chứng minh:

Giả sử A là tập con của E và cũng là tập con của tập cạnh của CKNN của G, và C là một thành phần liên thông trong rừng $F = (V, A)$. Nếu (u, v) là cạnh nhẹ nối C với một tplt khác trong F, thì (u, v) là an toàn đối với A.

Prim Algorithm - implementation

- Suppose the graph is given by a weight matrix $C=\{c(i,j), i, j = 1, 2,..., n\}$.
- Label of a node $v \in V \setminus V(T)$ is $(d[v], near[v])$:
  - $d[v]$ distance from the node v to $V(T)$:

    $d[v] := \min\{ c(v, w) : w \in V(T) \}$
  - $near[v] := z$ the nearest node to v (c(v, z) = d[v])



d(v) = 2
near[v] = w_5

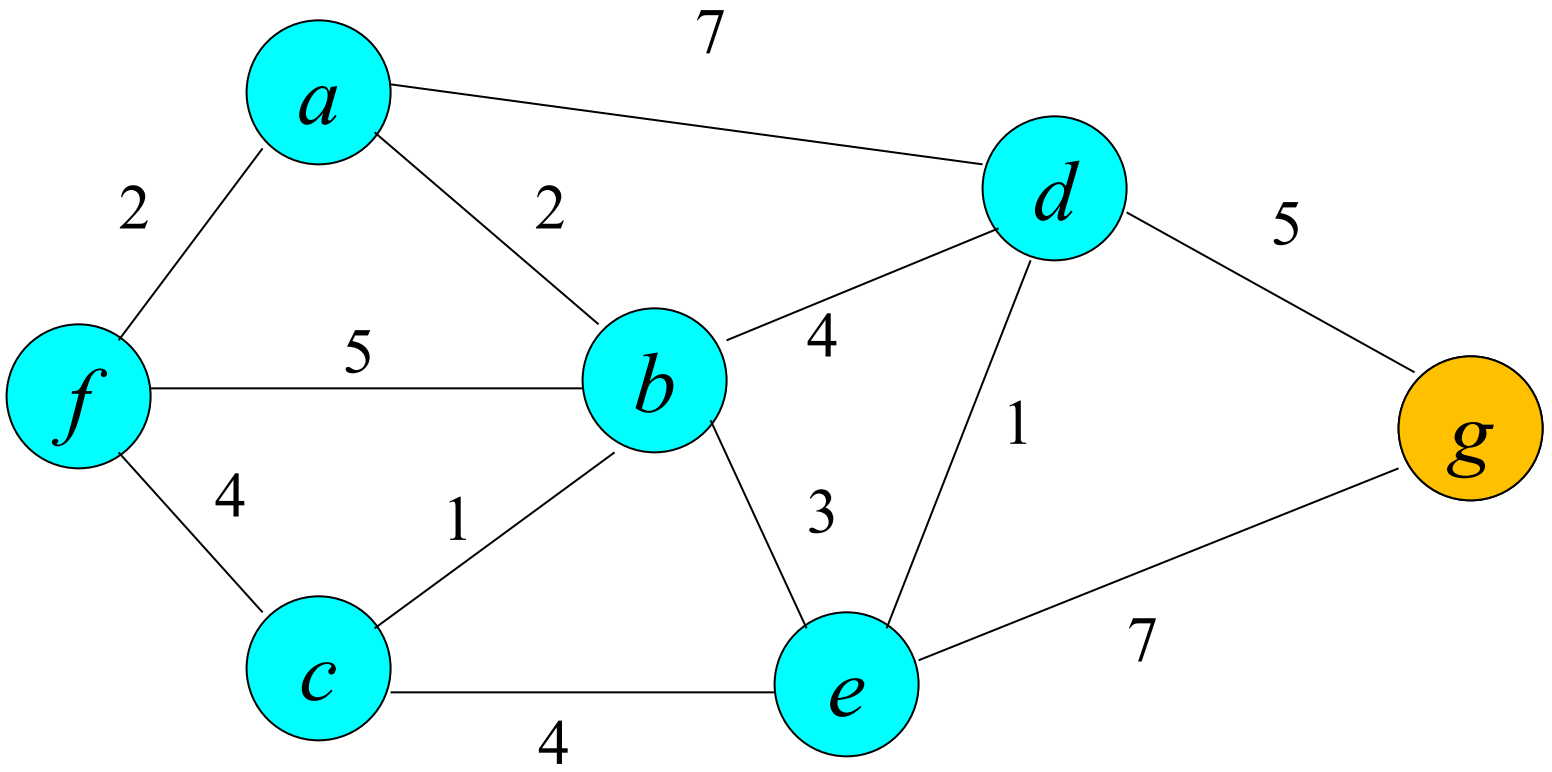
Prim Algorithm - implementation

```
Prim(G = (V, E)){
     Select r ∈ V; V(T)  = r; E(T) = ∅; d[r] = 0; near[r] = r;
     for v ∈ V \ {r} do   d[v] = c(r,v); near[v] = r;
     for k = 2 → n do {
          Select v ∈ V \ V(T) such that d[v] is minimal;
          V(T) = V(T) ∪ {v}; E(T) = E(T) ∪ {(v, near[v])};
          for u ∈ V \ V(T) do {
               if d[u] > c(v, u) then  { d[u] = c(v, u); near[u] = v;  }
          }
     }
     return (V(T), E(T));
}
```

Time complexity: $O(|V|^2)$

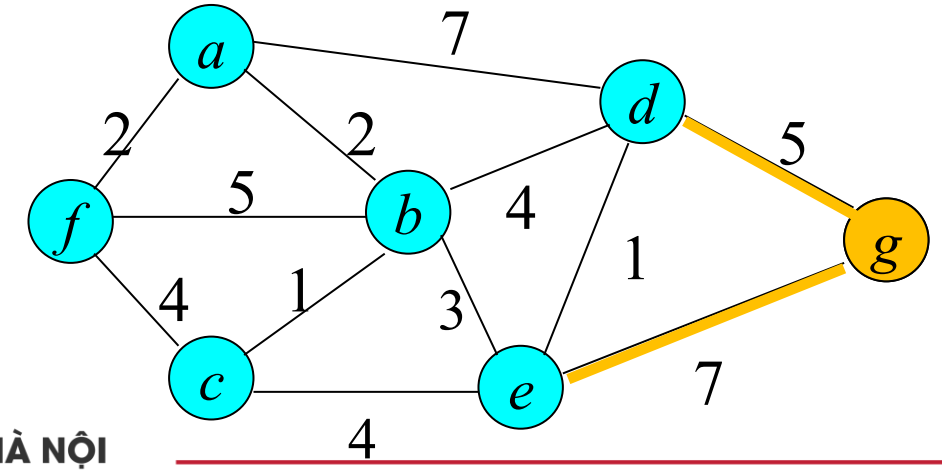
Prim Algorithm - implementation

- Example: Demonstrate the Prim algorithm for finding a minimum spanning tree of the graph below



Prim Algorithm - implementation

Node a	Node b	Node c	Node d	Node e	Node f	Node g	*V*(*T*)



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THANK YOU !